Printed Pages:02 **Subject Code:NEC-303**

Paper Id: 130316 **Roll No.** | | | | | | | | | | |

B TECH

(SEM-III) THEORY EXAMINATION 2018-19

SIGNALS AND SYSTEMS

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a) What is the difference between energy and power signal?

b) Sketch the following signal $x(t) = e^{-|t|}, -4 \le t \le 4$?

c) If $X(s) = e^{-2s}$ is the one sided Laplace transform of x (t), what is x (t)?

d) Find the unilateral Z-transform of $n^2u(n)$?

e) What is Fourier transform of $x(t) = \cos(\omega_c t + \Theta)$?

f) What is the relationship between LT and FT?

g) Show that the system y (n) = x (n) + y (n-1) is unstable?

h) What is the condition on the impulse response function of an LSI system if the system is to be memoryless?

i) Define poles and zeros?

j) Explain why an ideal LPF is not physically realizable?

SECTION B

2. Attempt any *three* of the following: **10 x 3 = 30**

a) Check whether the signals are periodic? If yes, find the fundamental period.

$$1.x(t) = a\sin 4t + b\cos 7t$$

$$2.x(n) = 2\cos^2 \pi n/6$$

b) Find the two-sided inverse Laplace transform of

$$X(s) = \frac{-3}{(s-1)(s+2)}, ROC - 2 < \sigma < 1?$$

c) Compare between CTFT & DTFT?

d) A discrete time system is described by the follow relation y(n)=nx(n)? Is this system (1) linear (2) time-invariant.

e) Obtain Cascade & Parallel realization of the transfer function $H(s) = \frac{5s^3}{s^3 + 6s^2 + 11s + 6}$?

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**

a) Sketch the following signal and determine their even and odd component

$$(1).x(t) = \begin{cases} 0.8t, 0 \le t \le 4 \\ 0, otherwise \end{cases}$$

$$(2).x(n) = \begin{cases} n, 1 \le n \le 3 \\ 0, otherwise \end{cases}$$

b) Determine the linear convolution of the sequence if x(n) = {1,-1,2} and y(n) = {1,0,1}.

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4. Attempt any *one* part of the following: **10 x 1 = 10**

a) Discuss Bilateral Z-transform & the properties of ROC?

b) Solve the First order difference equation x(n)-3x(n-1) = 6 with x (-1) = 4 using unilateral Z-transform.

5. Attempt any *one* part of the following: **10 x 1 = 10**

a) Sate and prove the Parseval's Theorem?

b) Determine the DTFT of $x(n)=(a^n \text{Cos}\omega n)U(n)$, $|a|<1$?

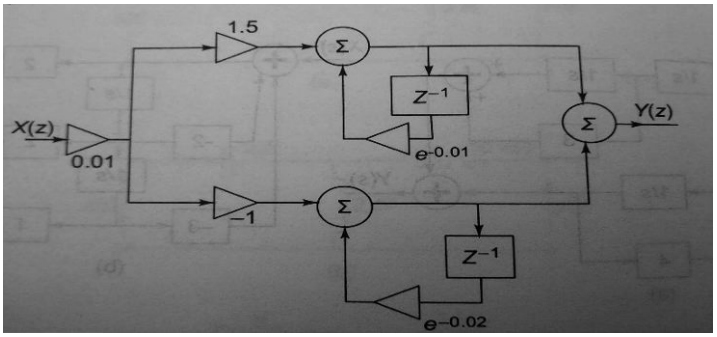
6. Attempt any *one* part of the following: **10 x 1 = 10**

a) Find the auto-correlation function and the energy spectral density of the signal $x(t) = A\Pi(t/2T)$?

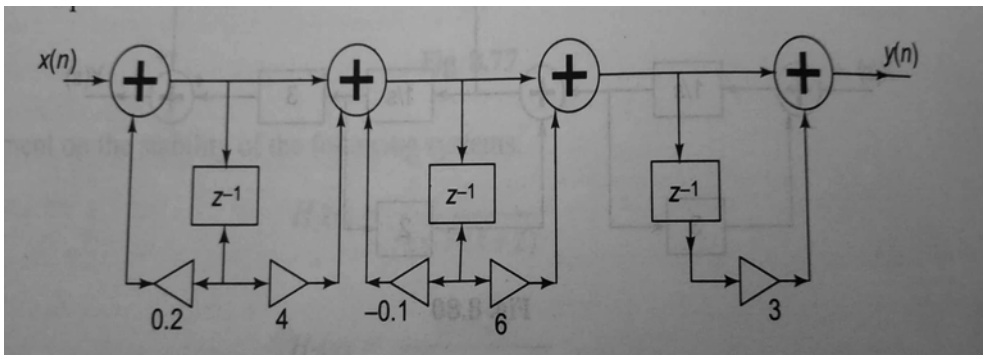
b) Show that the system given by the difference equation $y(n) = 1/4x(n) + 1/4y(n-1) - 1/2y(n-2)$ is causal and BIBO stable?

7. Attempt any *one* part of the following: **10 x 1 = 10**

a) Determine the system function H(z) and write down the difference equation of the system?



b) Determine the system function H(z) of the system shown & obtain its parallel realization?



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